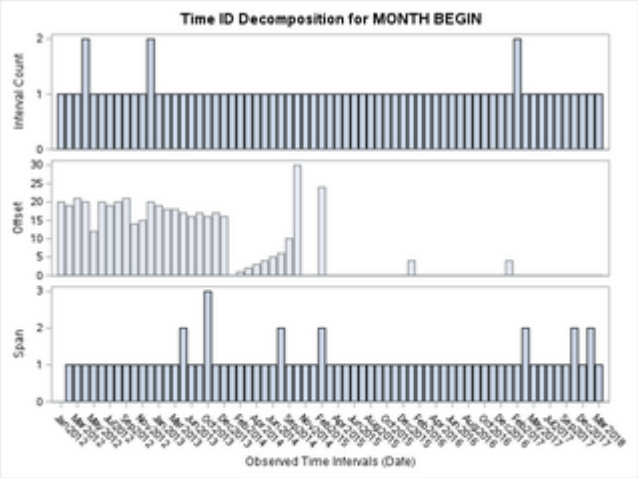
The TIMOID Procedure

| Input Data Set | |
|---|---|
| Name | WORK.PLAYSTORE |
| Label | |
| Time ID Variable | Date |
| Time Interval | MONTH |

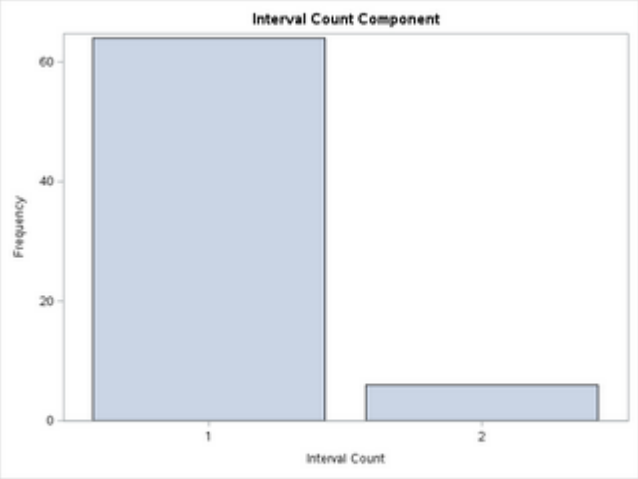
The TIMOID Procedure

| Time Component | | | | |
|---|---|---|---|---|
| Value Index | Date | Offset | Span | Interval Count |
| 1 | JAN2012 | 20 | . | 1 |
| 2 | FEB2012 | 19 | 1 | 1 |
| 3 | MAR2012 | 21 | 1 | 1 |
| 4 | APR2012 | 20 | 1 | 2 |
| 5 | MAY2012 | 12 | 1 | 1 |
| 6 | JUN2012 | 20 | 1 | 1 |
| 7 | JUL2012 | 19 | 1 | 1 |
| 8 | AUG2012 | 20 | 1 | 1 |
| 9 | SEP2012 | 21 | 1 | 1 |
| 10 | OCT2012 | 14 | 1 | 1 |
| 11 | NOV2012 | 15 | 1 | 1 |
| 12 | DEC2012 | 20 | 1 | 2 |
| 13 | JAN2013 | 19 | 1 | 1 |
| 14 | FEB2013 | 18 | 1 | 1 |
| 15 | MAR2013 | 18 | 1 | 1 |
| 16 | MAY2013 | 17 | 2 | 1 |
| 17 | JUN2013 | 16 | 1 | 1 |
| 18 | JUL2013 | 17 | 1 | 1 |
| 19 | OCT2013 | 16 | 3 | 1 |
| 20 | NOV2013 | 17 | 1 | 1 |
| 21 | DEC2013 | 16 | 1 | 1 |
| 22 | JAN2014 | 0 | 1 | 1 |
| 23 | FEB2014 | 1 | 1 | 1 |
| 24 | MAR2014 | 2 | 1 | 1 |
| 25 | APR2014 | 3 | 1 | 1 |
| 26 | MAY2014 | 4 | 1 | 1 |
| 27 | JUN2014 | 5 | 1 | 1 |
| 28 | AUG2014 | 6 | 2 | 1 |
| 29 | SEP2014 | 10 | 1 | 1 |
| 30 | OCT2014 | 30 | 1 | 1 |
| 31 | NOV2014 | 0 | 1 | 1 |
| 32 | DEC2014 | 0 | 1 | 1 |
| 33 | FEB2015 | 24 | 2 | 1 |
| 34 | MAR2015 | 0 | 1 | 1 |
| 35 | APR2015 | 0 | 1 | 1 |
| 36 | MAY2015 | 0 | 1 | 1 |
| 37 | JUN2015 | 0 | 1 | 1 |
| 38 | JUL2015 | 0 | 1 | 1 |
| 39 | AUG2015 | 0 | 1 | 1 |
| 40 | SEP2015 | 0 | 1 | 1 |
| 41 | OCT2015 | 0 | 1 | 1 |
| 42 | NOV2015 | 0 | 1 | 1 |
| 43 | DEC2015 | 0 | 1 | 1 |
| 44 | JAN2016 | 4 | 1 | 1 |
| 45 | FEB2016 | 0 | 1 | 1 |
| 46 | MAR2016 | 0 | 1 | 1 |
| 47 | APR2016 | 0 | 1 | 1 |
| 48 | MAY2016 | 0 | 1 | 1 |
| 49 | JUN2016 | 0 | 1 | 1 |
| 50 | JUL2016 | 0 | 1 | 1 |
| 51 | AUG2016 | 0 | 1 | 1 |
| 52 | SEP2016 | 0 | 1 | 1 |
| 53 | OCT2016 | 0 | 1 | 1 |
| 54 | NOV2016 | 0 | 1 | 1 |
| 55 | DEC2016 | 0 | 1 | 1 |
| 56 | JAN2017 | 4 | 1 | 1 |
| 57 | FEB2017 | 0 | 1 | 2 |
| 58 | APR2017 | 0 | 2 | 1 |
| 59 | MAY2017 | 0 | 1 | 1 |
| 60 | JUN2017 | 0 | 1 | 1 |
| 61 | JUL2017 | 0 | 1 | 1 |
| 62 | AUG2017 | 0 | 1 | 1 |
| 63 | SEP2017 | 0 | 1 | 1 |
| 64 | NOV2017 | 0 | 2 | 1 |
| 65 | DEC2017 | 0 | 1 | 1 |
| 66 | FEB2018 | 0 | 2 | 1 |
| 67 | MAR2018 | 0 | 1 | 1 |




The TIMOID Procedure

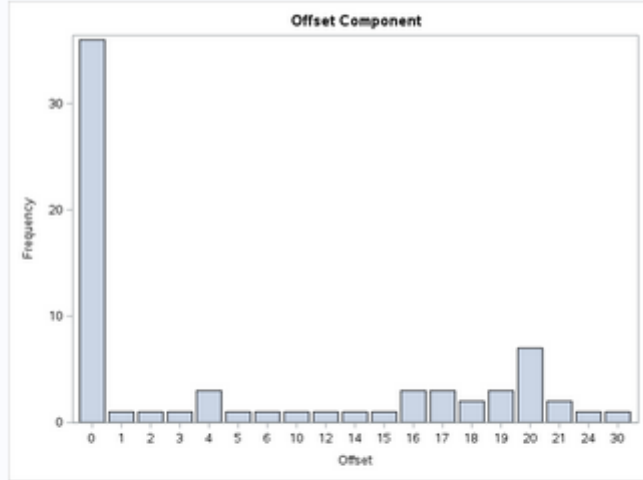
| Component | | | |
|---|---|---|---|
| Value Index | Interval Count | Frequency | Percentage |
| 1 | 1 | 64 | 91.428571 |
| 2 | 2 | 6 | 8.571429 |




| Statistics Summary | | | |
|---|---|---|---|
| Minimum | Maximum | Mean | Standard Deviation |
| 1 | 2 | 1.0857143 | 0.93395 |

The TIMOID Procedure

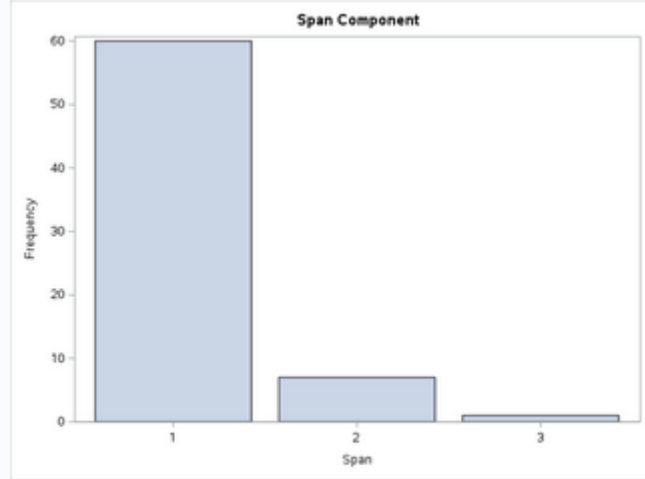
| Component | | | |
|---|---|---|---|
| Value Index | Offset | Frequency | Percentage |
| 1 | 0 | 36 | 51.428571 |
| 2 | 1 | 1 | 1.428571 |
| 3 | 2 | 1 | 1.428571 |
| 4 | 3 | 1 | 1.428571 |
| 5 | 4 | 3 | 4.285714 |
| 6 | 5 | 1 | 1.428571 |
| 7 | 6 | 1 | 1.428571 |
| 8 | 10 | 1 | 1.428571 |
| 9 | 12 | 1 | 1.428571 |
| 10 | 14 | 1 | 1.428571 |
| 11 | 15 | 1 | 1.428571 |
| 12 | 16 | 3 | 4.285714 |
| 13 | 17 | 3 | 4.285714 |
| 14 | 18 | 2 | 2.857143 |
| 15 | 19 | 3 | 4.285714 |
| 16 | 20 | 7 | 10 |
| 17 | 21 | 2 | 2.857143 |
| 18 | 24 | 1 | 1.428571 |
| 19 | 30 | 1 | 1.428571 |




| Statistics Summary | | | |
|---|---|---|---|
| Minimum | Maximum | Mean | Standard Deviation |
| 0 | 30 | 7.2571429 | 34.332644 |

The TIMOID Procedure

| Component | | | |
|---|---|---|---|
| Value Index | Span | Frequency | Percentage |
| 1 | 1 | 60 | 85.715294 |
| 2 | 2 | 7 | 10.294118 |
| 3 | 3 | 1 | 1.470588 |

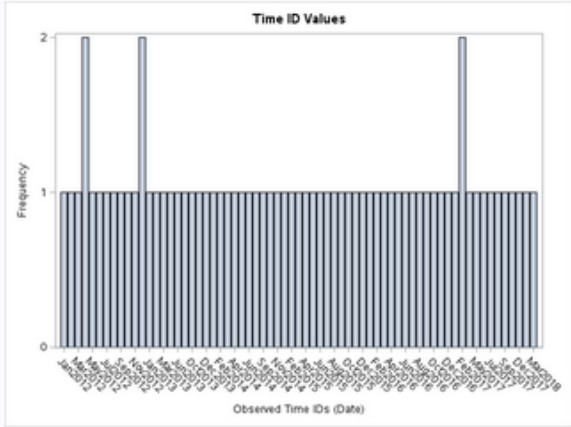



| Statistics Summary | | | |
|---|---|---|---|
| Minimum | Maximum | Mean | Standard Deviation |
| 1 | 3 | 1.1323529 | 1.2425188 |

The TIMEID Procedure

| Time ID Values for Date | | | |
|---|---|---|---|
| Value Index | Date | Frequency | Percentage |
| 1 | JAN2012 | 1 | 1.428571 |
| 2 | FEB2012 | 1 | 1.428571 |
| 3 | MAR2012 | 1 | 1.428571 |
| 4 | APR2012 | 2 | 2.857143 |
| 5 | MAY2012 | 1 | 1.428571 |
| 6 | JUN2012 | 1 | 1.428571 |
| 7 | JUL2012 | 1 | 1.428571 |
| 8 | AUG2012 | 1 | 1.428571 |
| 9 | SEP2012 | 1 | 1.428571 |
| 10 | OCT2012 | 1 | 1.428571 |
| 11 | NOV2012 | 1 | 1.428571 |
| 12 | DEC2012 | 2 | 2.857143 |
| 13 | JAN2013 | 1 | 1.428571 |
| 14 | FEB2013 | 1 | 1.428571 |
| 15 | MAR2013 | 1 | 1.428571 |
| 16 | MAY2013 | 1 | 1.428571 |
| 17 | JUN2013 | 1 | 1.428571 |
| 18 | JUL2013 | 1 | 1.428571 |
| 19 | OCT2013 | 1 | 1.428571 |
| 20 | NOV2013 | 1 | 1.428571 |
| 21 | DEC2013 | 1 | 1.428571 |
| 22 | JAN2014 | 1 | 1.428571 |
| 23 | FEB2014 | 1 | 1.428571 |
| 24 | MAR2014 | 1 | 1.428571 |
| 25 | APR2014 | 1 | 1.428571 |
| 26 | MAY2014 | 1 | 1.428571 |
| 27 | JUN2014 | 1 | 1.428571 |
| 28 | AUG2014 | 1 | 1.428571 |
| 29 | SEP2014 | 1 | 1.428571 |
| 30 | OCT2014 | 1 | 1.428571 |
| 31 | NOV2014 | 1 | 1.428571 |
| 32 | DEC2014 | 1 | 1.428571 |
| 33 | FEB2015 | 1 | 1.428571 |
| 34 | MAR2015 | 1 | 1.428571 |
| 35 | APR2015 | 1 | 1.428571 |
| 36 | MAY2015 | 1 | 1.428571 |
| 37 | JUN2015 | 1 | 1.428571 |
| 38 | JUL2015 | 1 | 1.428571 |
| 39 | AUG2015 | 1 | 1.428571 |
| 40 | SEP2015 | 1 | 1.428571 |
| 41 | OCT2015 | 1 | 1.428571 |
| 42 | NOV2015 | 1 | 1.428571 |
| 43 | DEC2015 | 1 | 1.428571 |
| 44 | JAN2016 | 1 | 1.428571 |
| 45 | FEB2016 | 1 | 1.428571 |
| 46 | MAR2016 | 1 | 1.428571 |
| 47 | APR2016 | 1 | 1.428571 |
| 48 | MAY2016 | 1 | 1.428571 |
| 49 | JUN2016 | 1 | 1.428571 |
| 50 | JUL2016 | 1 | 1.428571 |
| 51 | AUG2016 | 1 | 1.428571 |
| 52 | SEP2016 | 1 | 1.428571 |
| 53 | OCT2016 | 1 | 1.428571 |
| 54 | NOV2016 | 1 | 1.428571 |
| 55 | DEC2016 | 1 | 1.428571 |
| 56 | JAN2017 | 1 | 1.428571 |
| 57 | FEB2017 | 2 | 2.857143 |
| 58 | APR2017 | 1 | 1.428571 |
| 59 | MAY2017 | 1 | 1.428571 |
| 60 | JUN2017 | 1 | 1.428571 |
| 61 | JUL2017 | 1 | 1.428571 |
| 62 | AUG2017 | 1 | 1.428571 |
| 63 | SEP2017 | 1 | 1.428571 |
| 64 | NOV2017 | 1 | 1.428571 |
| 65 | DEC2017 | 1 | 1.428571 |
| 66 | FEB2018 | 1 | 1.428571 |
| 67 | MAR2018 | 1 | 1.428571 |

| Time ID Value Summary for Date | | | | |
|---|---|---|---|---|
| N Obs Excluded | N Obs Read | N Obs Missing | N Obs Used | Unique Values |
| 0 | 70 | 0 | 70 | 67 |




| Time Interval Analysis Summary | |
|---|---|
| Time ID Variable | Date |
| Time Interval | MONTH |
| Base Name | MONTH |
| Multiplier | 1 |
| Shift | 0 |
| Length of Seasonal Cycle | 12 |
| Time ID Format | MONYY7. |
| Start | JAN2012 |
| End | MAR2018 |

The CONTENTS Procedure

| | | | |
|---|---|---|---|
| Data Set Name | WORK.INFLATION | Observations | 75 |
| Member Type | DATA | Variables | 11 |
| Engine | V9 | Indexes | 0 |
| Created | 04/05/2026 13:04:22 | Observation Length | 88 |
| Last Modified | 04/05/2026 13:04:22 | Deleted Observations | 0 |
| Protection | | Compressed | NO |
| Data Set Type | | Sorted | NO |
| Label | | | |
| Data Representation | SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64 | | |
| Encoding | utf-8 Unicode (UTF-8) | | |

| Engine/Host Dependent Information | |
|---|---|
| Data Set Page Size | 131072 |
| Number of Data Set Pages | 1 |
| First Data Page | 1 |
| Max Obs per Page | 1486 |
| Obs In First Data Page | 75 |
| Number of Data Set Repairs | 0 |
| Filename | /saswork/SAS_work3580000189C7E_odaws01-usw2.oda.sas.com/SAS_work60D9000189C7E_odaws01-usw2.oda.sas.com/inflation.sas7bdat |
| Release Created | 9.0401M8 |
| Host Created | Linux |
| Inode Number | 1874847254 |
| Access Permission | rw-r--r-- |
| Owner Name | u64360180 |
| File Size | 256KB |
| File Size (bytes) | 262144 |

| Alphabetic List of Variables and Attributes | | | | | |
|---|---|---|---|---|---|
| # | Variable | Type | Len | Format | Label |
| 11 | CPI | Num | 8 | BEST. | CPI |
| 8 | Clothing_and_shopping | Num | 8 | BEST. | Clothing_and_shopping |
| 6 | Communication | Num | 8 | BEST. | Communication |
| 4 | Eating_out | Num | 8 | BEST. | Eating_out |
| 7 | Education | Num | 8 | BEST. | Education |
| 5 | Entertainment | Num | 8 | BEST. | Entertainment |
| 3 | Furniture_Home_Improvement | Num | 8 | BEST. | Furniture_Home_Improvement |
| 9 | Grocery | Num | 8 | BEST. | Grocery |
| 1 | Month | Num | 8 | MONYY7. | Month |
| 10 | Spend_save_quaterly_ratio | Num | 8 | BEST. | Spend_save_quaterly_ratio |
| 2 | Travel_including_Leisure | Num | 8 | BEST. | Travel_including_Leisure |

*Program 1 ×  *Program 2 ×

CODE  LOG  RESULTS  OUTPUT DATA

Table: WORK.MODEL  View: Column names  Filter: (none)

Columns

- [x] Select all
- [x] Month
- [x] Furniture_Home_Improvement
- [x] Travel_including_Leisure
- [x] Eating_out
- [x] Entertainment
- [x] Grocery
- [x] Education
- [x] Communication
- [x] Clothing_and_shopping
- [x] Spend_save_quaterly_ratio
- [x] CPI

| Property | Value |
|---|---|
| Label | |
| Name | |
| Length | |
| Type | |
| Format | |
| Informat | |

Total rows: 75  Total columns: 11  Rows 1-75

| | Month | Furniture_Home_Improvement | Travel_including_Leisure | Eating_out | Entertainment |
|---|---|---|---|---|---|
| 1 | JAN2012 | 100 | 100 | 100 | 100 |
| 2 | FEB2012 | 101 | 101 | 100 | 101 |
| 3 | MAR2012 | 104 | 101 | 100 | 103 |
| 4 | APR2012 | 106 | 101 | 100 | 105 |
| 5 | MAY2012 | 110 | 103 | 102 | 105 |
| 6 | JUN2012 | 111 | 105 | 102 | 105 |
| 7 | JUL2012 | 112 | 105 | 102 | 105 |
| 8 | AUG2012 | 112 | 106 | 101 | 106 |
| 9 | SEP2012 | 112 | 107 | 101 | 106 |
| 10 | OCT2012 | 114 | 107 | 102 | 107 |
| 11 | NOV2012 | 113 | 107 | 102 | 108 |
| 12 | DEC2012 | 113 | 108 | 103 | 109 |
| 13 | JAN2013 | 112 | 108 | 102 | 110 |
| 14 | FEB2013 | 113 | 109 | 100 | 111 |
| 15 | MAR2013 | 113 | 110 | 100 | 112 |
| 16 | APR2013 | 114 | 111 | 100 | 112 |
| 17 | MAY2013 | 115 | 114 | 102 | 112 |
| 18 | JUN2013 | 116 | 115 | 103 | 112 |
| 19 | JUL2013 | 115 | 116 | 103 | 112 |
| 20 | AUG2013 | 115 | 116 | 104 | 112 |
| 21 | SEP2013 | 115 | 116 | 105 | 112 |
| 22 | OCT2013 | 115 | 116 | 105 | 113 |
| 23 | NOV2013 | 115 | 116 | 104 | 112 |
| 24 | DEC2013 | 116 | 116 | 103 | 111 |
| 25 | JAN2014 | 116 | 117 | 103 | 112 |

Messages: 3  User: u64360188

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

| | |
|---|---|
| Number of Observations Read | 75 |
| Number of Observations Used | 69 |
| Number of Observations with Missing Values | 6 |

Forward Selection: Step 1

| Statistics for Entry DF = 1,67 | | | | |
|---|---|---|---|---|
| Variable | Tolerance | Model R-Square | F Value | Pr > F |
| Furniture_Home_Improvement | 1.000000 | 0.2405 | 21.21 | <.0001 |
| Travel_Including_Leisure | 1.000000 | 0.5712 | 89.27 | <.0001 |
| Eating_out | 1.000000 | 0.6670 | 134.20 | <.0001 |
| Entertainment | 1.000000 | 0.4943 | 65.50 | <.0001 |
| Grocery | 1.000000 | 0.6629 | 131.73 | <.0001 |
| Education | 1.000000 | 0.8195 | 304.24 | <.0001 |
| Communication | 1.000000 | 0.0281 | 1.93 | 0.1689 |
| Clothing_and_shopping | 1.000000 | 0.8239 | 313.54 | <.0001 |
| Spend_save_quaterly_ratio | 1.000000 | 0.8661 | 433.27 | <.0001 |

Variable Spend_save_quaterly_ratio Entered: R-Square = 0.8661 and C(p) = 101.1809

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 1 | 18.25606 | 18.25606 | 433.27 | <.0001 |
| Error | 67 | 2.82307 | 0.04214 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 91.90309 | 0.64805 | 847.39683 | 20111.3 | <.0001 |
| Spend_save_quaterly_ratio | 0.12923 | 0.00621 | 18.25606 | 433.27 | <.0001 |

Bounds on condition number: 1, 1

Forward Selection: Step 2

| Statistics for Entry DF = 1,66 | | | | |
|---|---|---|---|---|
| Variable | Tolerance | Model R-Square | F Value | Pr > F |
| Furniture_Home_Improvement | 0.689061 | 0.8673 | 0.59 | 0.4457 |
| Travel_Including_Leisure | 0.347601 | 0.8661 | 0.02 | 0.8770 |
| Eating_out | 0.265277 | 0.8674 | 0.68 | 0.4133 |
| Entertainment | 0.355325 | 0.8716 | 2.82 | 0.0980 |
| Grocery | 0.228051 | 0.8661 | 0.03 | 0.8715 |
| Education | 0.148517 | 0.8806 | 8.06 | 0.0060 |
| Communication | 0.944481 | 0.8689 | 1.43 | 0.2362 |
| Clothing_and_shopping | 0.240857 | 0.9050 | 27.07 | <.0001 |

Variable Clothing_and_shopping Entered: R-Square = 0.9050 and C(p) = 54.8481

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 2 | 19.07714 | 9.53857 | 314.46 | <.0001 |
| Error | 66 | 2.00199 | 0.03033 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 85.53920 | 1.34109 | 123.40594 | 4068.34 | <.0001 |
| Clothing_and_shopping | 0.11089 | 0.02131 | 0.82107 | 27.07 | <.0001 |
| Spend_save_quaterly_ratio | 0.08058 | 0.01073 | 1.70936 | 56.35 | <.0001 |

Bounds on condition number: 4.1518, 16.607

Forward Selection: Step 3

| Statistics for Entry DF = 1,65 | | | | |
|---|---|---|---|---|
| Variable | Tolerance | Model R-Square | F Value | Pr > F |
| Furniture_Home_Improvement | 0.481411 | 0.9342 | 28.81 | <.0001 |
| Travel_Including_Leisure | 0.198806 | 0.9311 | 24.60 | <.0001 |
| Eating_out | 0.247564 | 0.9052 | 0.15 | 0.7035 |
| Entertainment | 0.285849 | 0.9374 | 33.57 | <.0001 |
| Grocery | 0.186638 | 0.9152 | 7.83 | 0.0068 |
| Education | 0.068038 | 0.9063 | 0.91 | 0.3428 |
| Communication | 0.940321 | 0.9066 | 1.13 | 0.2920 |

Variable Entertainment Entered: R-Square = 0.9374 and C(p) = 16.7114

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 3 | 19.75898 | 6.58633 | 324.29 | <.0001 |
| Error | 65 | 1.32016 | 0.02031 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 83.43388 | 1.15596 | 105.80571 | 5209.52 | <.0001 |
| Entertainment | -0.04118 | 0.00711 | 0.68184 | 33.57 | <.0001 |
| Clothing_and_shopping | 0.16070 | 0.01944 | 1.38735 | 68.31 | <.0001 |
| Spend_save_quaterly_ratio | 0.09622 | 0.00919 | 2.22716 | 109.66 | <.0001 |

Bounds on condition number: 5.161, 39.611

Forward Selection: Step 4

| Statistics for Entry DF = 1,64 | | | | |
|---|---|---|---|---|
| Variable | Tolerance | Model R-Square | F Value | Pr > F |
| Furniture_Home_Improvement | 0.127589 | 0.9384 | 1.08 | 0.3026 |
| Travel_Including_Leisure | 0.050971 | 0.9375 | 0.16 | 0.6873 |
| Eating_out | 0.239789 | 0.9377 | 0.32 | 0.5757 |
| Grocery | 0.101851 | 0.9381 | 0.77 | 0.3833 |
| Education | 0.036368 | 0.9513 | 18.38 | <.0001 |
| Communication | 0.938853 | 0.9385 | 1.15 | 0.2885 |

Variable Education Entered: R-Square = 0.9513 and C(p) = 1.3699

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 4 | 20.05357 | 5.01339 | 312.86 | <.0001 |
| Error | 64 | 1.02556 | 0.01602 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 84.33993 | 1.04830 | 103.72301 | 6472.84 | <.0001 |
| Entertainment | -0.06644 | 0.00863 | 0.94870 | 59.20 | <.0001 |
| Education | 0.11236 | 0.02620 | 0.29460 | 18.38 | <.0001 |
| Clothing_and_shopping | 0.09246 | 0.02349 | 0.24830 | 15.50 | 0.0002 |
| Spend_save_quaterly_ratio | 0.06974 | 0.01024 | 0.74399 | 46.43 | <.0001 |

Bounds on condition number: 27.497, 202.93

Forward Selection: Step 5

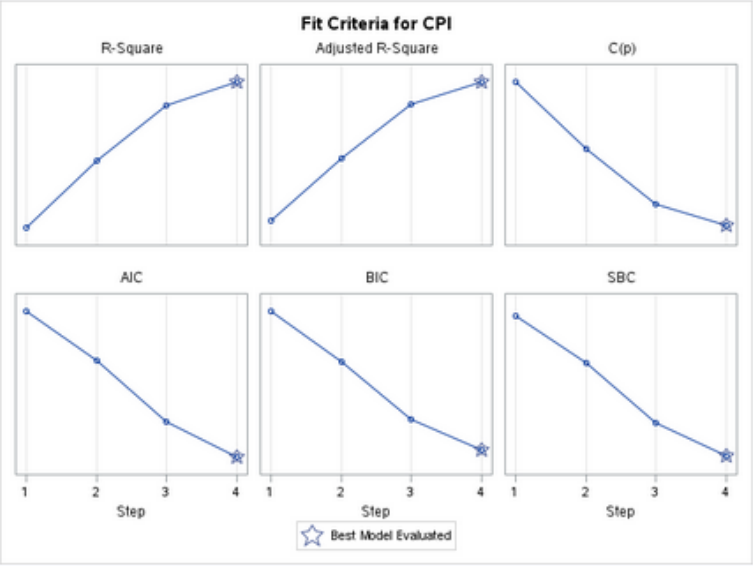
| Statistics for Entry DF = 1,63 | | | | |
|---|---|---|---|---|
| Variable | Tolerance | Model R-Square | F Value | Pr > F |
| Furniture_Home_Improvement | 0.105777 | 0.9517 | 0.44 | 0.5117 |
| Travel_Including_Leisure | 0.050890 | 0.9516 | 0.39 | 0.5330 |
| Eating_out | 0.239778 | 0.9516 | 0.37 | 0.5477 |
| Grocery | 0.096775 | 0.9513 | 0.00 | 0.9734 |
| Communication | 0.918195 | 0.9516 | 0.33 | 0.5701 |

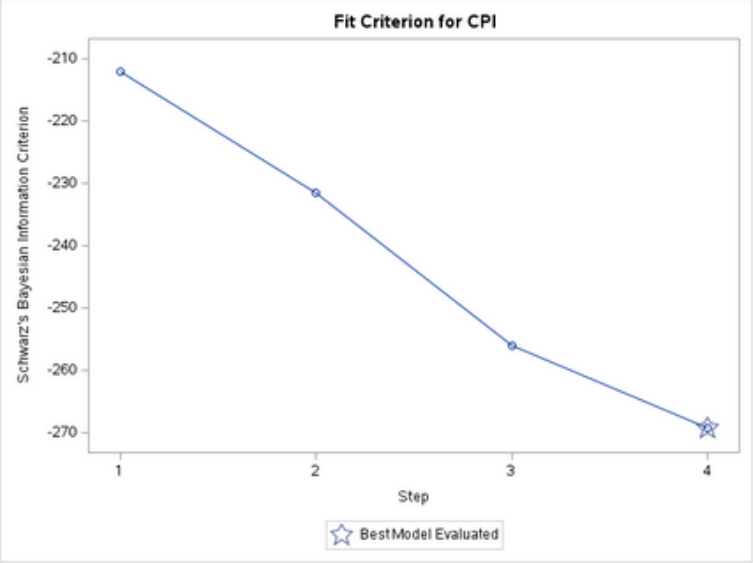
No other variable met the 0.5000 significance level for entry into the model.

Summary of Forward Selection

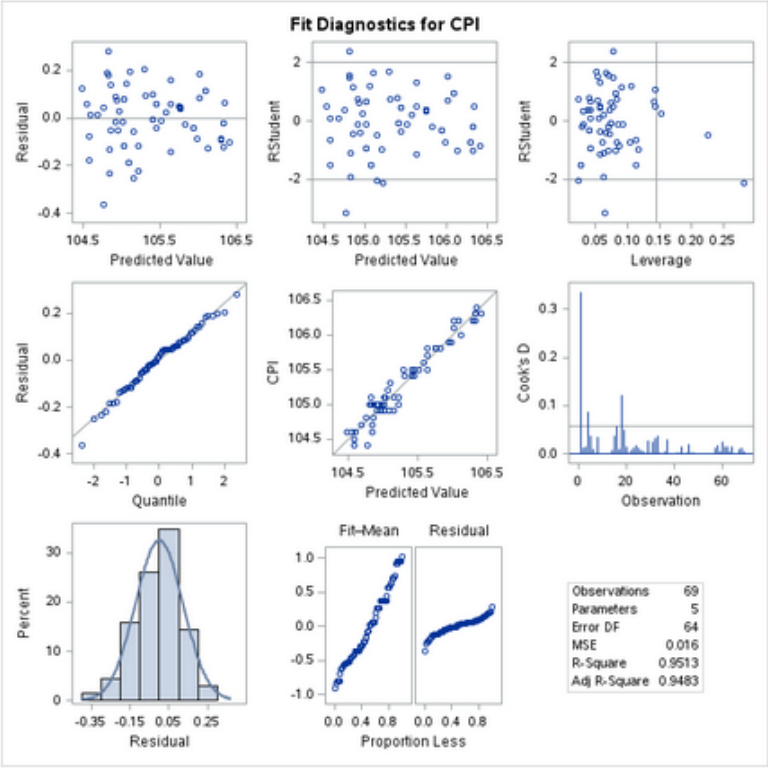
| Step | Variable Entered | Label | Number Vars In | Partial R-Square | Model R-Square | C(p) | F Value | Pr > F |
|---|---|---|---|---|---|---|---|---|
| 1 | Spend_save_quaterly_ratio | Spend_save_quaterly_ratio | 1 | 0.8661 | 0.8661 | 101.181 | 433.27 | <.0001 |
| 2 | Clothing_and_shopping | Clothing_and_shopping | 2 | 0.0390 | 0.9050 | 54.8481 | 27.07 | <.0001 |
| 3 | Entertainment | Entertainment | 3 | 0.0323 | 0.9374 | 16.7114 | 33.57 | <.0001 |
| 4 | Education | Education | 4 | 0.0140 | 0.9513 | 1.3699 | 18.38 | <.0001 |

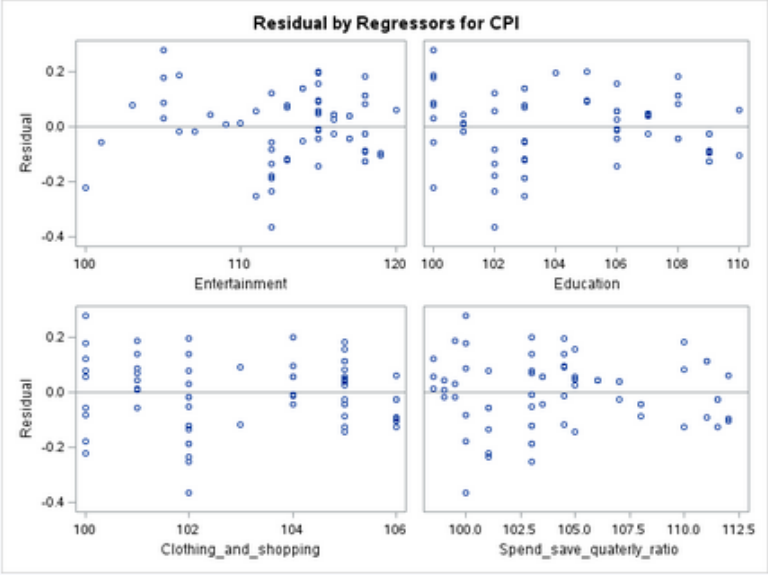
The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI




The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI




The REG Procedure
Model: MODEL2
Dependent Variable: CPI CPI

| | |
|---|---|
| Number of Observations Read | 75 |
| Number of Observations Used | 69 |
| Number of Observations with Missing Values | 6 |

Backward Elimination: Step 0

All Variables Entered: R-Square = 0.9525 and C(p) = 10.0000

Analysis of Variance

| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
|---|---|---|---|---|---|
| Model | 9 | 20.07684 | 2.23076 | 131.31 | <.0001 |
| Error | 59 | 1.00229 | 0.01699 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 83.13330 | 3.04430 | 12.66819 | 745.72 | <.0001 |
| Furniture_Home_Improvement | 0.00911 | 0.01382 | 0.00738 | 0.43 | 0.5125 |
| Travel_Including_Leisure | -0.00750 | 0.01098 | 0.00793 | 0.47 | 0.4972 |
| Eating_out | -0.00015179 | 0.01061 | 0.00000348 | 0.00 | 0.9886 |
| Entertainment | -0.06756 | 0.01989 | 0.19602 | 11.54 | 0.0012 |
| Grocery | 0.00123 | 0.03527 | 0.00002060 | 0.00 | 0.9723 |
| Education | 0.11989 | 0.03200 | 0.23849 | 14.04 | 0.0004 |
| Communication | -0.00094288 | 0.00152 | 0.00656 | 0.39 | 0.5367 |
| Clothing_and_shopping | 0.09299 | 0.03123 | 0.15059 | 8.86 | 0.0042 |
| Spend_save_quaterly_ratio | 0.07273 | 0.01410 | 0.45186 | 26.60 | <.0001 |

Bounds on condition number: 38.873, 1478

Backward Elimination: Step 1

Statistics for Removal
DF = 1,59

| Variable | Partial R-Square | Model R-Square | F Value | Pr > F |
|---|---|---|---|---|
| Furniture_Home_Improvement | 0.0004 | 0.9521 | 0.43 | 0.5125 |
| Travel_Including_Leisure | 0.0004 | 0.9521 | 0.47 | 0.4972 |
| Eating_out | 0.0000 | 0.9525 | 0.00 | 0.9886 |
| Entertainment | 0.0093 | 0.9432 | 11.54 | 0.0012 |
| Grocery | 0.0000 | 0.9525 | 0.00 | 0.9723 |
| Education | 0.0113 | 0.9411 | 14.04 | 0.0004 |
| Communication | 0.0003 | 0.9521 | 0.39 | 0.5367 |
| Clothing_and_shopping | 0.0071 | 0.9453 | 8.86 | 0.0042 |
| Spend_save_quaterly_ratio | 0.0214 | 0.9310 | 26.60 | <.0001 |

Variable Eating_out Removed: R-Square = 0.9525 and C(p) = 8.0002

Analysis of Variance

| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
|---|---|---|---|---|---|
| Model | 8 | 20.07684 | 2.50960 | 150.23 | <.0001 |
| Error | 60 | 1.00229 | 0.01670 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 83.14480 | 2.91172 | 13.62114 | 815.40 | <.0001 |
| Furniture_Home_Improvement | 0.00901 | 0.01205 | 0.00934 | 0.56 | 0.4576 |
| Travel_Including_Leisure | -0.00741 | 0.00878 | 0.01188 | 0.71 | 0.4024 |
| Entertainment | -0.06759 | 0.01959 | 0.19883 | 11.90 | 0.0010 |
| Grocery | 0.00130 | 0.03461 | 0.00002364 | 0.00 | 0.9701 |
| Education | 0.11976 | 0.03049 | 0.25767 | 15.42 | 0.0002 |
| Communication | -0.00094349 | 0.00150 | 0.00657 | 0.39 | 0.5328 |
| Clothing_and_shopping | 0.09292 | 0.03056 | 0.15446 | 9.25 | 0.0035 |
| Spend_save_quaterly_ratio | 0.07263 | 0.01205 | 0.60671 | 36.32 | <.0001 |

Bounds on condition number: 35.713, 1083.5

Backward Elimination: Step 2

| Statistics for Removal DF = 1,60 | | | | |
|---|---|---|---|---|
| Variable | Partial R-Square | Model R-Square | F Value | Pr > F |
| Furniture_Home_Improvement | 0.0004 | 0.9520 | 0.56 | 0.4576 |
| Travel_Including_Leisure | 0.0006 | 0.9519 | 0.71 | 0.4024 |
| Entertainment | 0.0094 | 0.9430 | 11.90 | 0.0010 |
| Grocery | 0.0000 | 0.9524 | 0.00 | 0.9701 |
| Education | 0.0122 | 0.9402 | 15.42 | 0.0002 |
| Communication | 0.0003 | 0.9521 | 0.39 | 0.5328 |
| Clothing_and_shopping | 0.0073 | 0.9451 | 9.25 | 0.0035 |
| Spend_save_quaterly_ratio | 0.0288 | 0.9237 | 36.32 | <.0001 |

Variable Grocery Removed: R-Square = 0.9524 and C(p) = 6.0016

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 7 | 20.07682 | 2.86812 | 174.55 | <.0001 |
| Error | 61 | 1.00231 | 0.01643 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 83.23078 | 1.78902 | 35.56396 | 2164.39 | <.0001 |
| Furniture_Home_Improvement | 0.00901 | 0.01195 | 0.00933 | 0.57 | 0.4539 |
| Travel_Including_Leisure | -0.00739 | 0.00869 | 0.01186 | 0.72 | 0.3988 |
| Entertainment | -0.06744 | 0.01901 | 0.20669 | 12.58 | 0.0008 |
| Education | 0.11998 | 0.02969 | 0.26828 | 16.33 | 0.0002 |
| Communication | -0.00094741 | 0.00149 | 0.00666 | 0.41 | 0.5267 |
| Clothing_and_shopping | 0.09286 | 0.03027 | 0.15465 | 9.41 | 0.0032 |
| Spend_save_quaterly_ratio | 0.07274 | 0.01155 | 0.65162 | 39.66 | <.0001 |

Bounds on condition number: 34.431, 860.97

Backward Elimination: Step 3

| Statistics for Removal DF = 1,61 | | | | |
|---|---|---|---|---|
| Variable | Partial R-Square | Model R-Square | F Value | Pr > F |
| Furniture_Home_Improvement | 0.0004 | 0.9520 | 0.57 | 0.4539 |
| Travel_Including_Leisure | 0.0006 | 0.9519 | 0.72 | 0.3988 |
| Entertainment | 0.0098 | 0.9426 | 12.58 | 0.0008 |
| Education | 0.0127 | 0.9397 | 16.33 | 0.0002 |
| Communication | 0.0003 | 0.9521 | 0.41 | 0.5267 |
| Clothing_and_shopping | 0.0073 | 0.9451 | 9.41 | 0.0032 |
| Spend_save_quaterly_ratio | 0.0309 | 0.9215 | 39.66 | <.0001 |

Variable Communication Removed: R-Square = 0.9521 and C(p) = 4.3937

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 6 | 20.07016 | 3.34503 | 205.55 | <.0001 |
| Error | 62 | 1.00898 | 0.01627 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 83.31267 | 1.77582 | 35.81913 | 2201.03 | <.0001 |
| Furniture_Home_Improvement | 0.00942 | 0.01188 | 0.01023 | 0.63 | 0.4310 |
| Travel_Including_Leisure | -0.00655 | 0.00855 | 0.00954 | 0.59 | 0.4467 |
| Entertainment | -0.06974 | 0.01858 | 0.22934 | 14.09 | 0.0004 |
| Education | 0.12287 | 0.02920 | 0.28805 | 17.70 | <.0001 |
| Clothing_and_shopping | 0.09037 | 0.02987 | 0.14894 | 9.15 | 0.0036 |
| Spend_save_quaterly_ratio | 0.07167 | 0.01137 | 0.64629 | 39.71 | <.0001 |

Bounds on condition number: 33.828, 706.41

Backward Elimination: Step 4

| Statistics for Removal DF = 1,62 | | | | |
|---|---|---|---|---|
| Variable | Partial R-Square | Model R-Square | F Value | Pr > F |
| Furniture_Home_Improvement | 0.0005 | 0.9516 | 0.63 | 0.4310 |
| Travel_Including_Leisure | 0.0005 | 0.9517 | 0.59 | 0.4467 |
| Entertainment | 0.0109 | 0.9413 | 14.09 | 0.0004 |
| Education | 0.0137 | 0.9385 | 17.70 | <.0001 |
| Clothing_and_shopping | 0.0071 | 0.9451 | 9.15 | 0.0036 |
| Spend_save_quaterly_ratio | 0.0307 | 0.9215 | 39.71 | <.0001 |

Variable Travel_Including_Leisure Removed: R-Square = 0.9517 and C(p) = 2.9554

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 5 | 20.06061 | 4.01212 | 248.17 | <.0001 |
| Error | 63 | 1.01852 | 0.01617 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 84.40323 | 1.05731 | 103.02395 | 6372.50 | <.0001 |
| Furniture_Home_Improvement | 0.00767 | 0.01162 | 0.00704 | 0.44 | 0.5117 |
| Entertainment | -0.07587 | 0.01671 | 0.33320 | 20.61 | <.0001 |
| Education | 0.12025 | 0.02891 | 0.27973 | 17.30 | <.0001 |
| Clothing_and_shopping | 0.08254 | 0.02797 | 0.14074 | 8.71 | 0.0044 |
| Spend_save_quaterly_ratio | 0.07275 | 0.01125 | 0.67641 | 41.84 | <.0001 |

Bounds on condition number: 33.187, 444.45

Backward Elimination: Step 5

| Statistics for Removal DF = 1,63 | | | | |
|---|---|---|---|---|
| Variable | Partial R-Square | Model R-Square | F Value | Pr > F |
| Furniture_Home_Improvement | 0.0003 | 0.9513 | 0.44 | 0.5117 |
| Entertainment | 0.0158 | 0.9359 | 20.61 | <.0001 |
| Education | 0.0133 | 0.9384 | 17.30 | <.0001 |
| Clothing_and_shopping | 0.0067 | 0.9450 | 8.71 | 0.0044 |
| Spend_save_quaterly_ratio | 0.0321 | 0.9196 | 41.84 | <.0001 |

Variable Furniture_Home_Improvement Removed: R-Square = 0.9513 and C(p) = 1.3699

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 4 | 20.05357 | 5.01339 | 312.86 | <.0001 |
| Error | 64 | 1.02556 | 0.01602 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 84.33993 | 1.04830 | 103.72301 | 6472.84 | <.0001 |
| Entertainment | -0.06644 | 0.00863 | 0.94870 | 59.20 | <.0001 |
| Education | 0.11236 | 0.02620 | 0.29460 | 18.38 | <.0001 |
| Clothing_and_shopping | 0.09246 | 0.02349 | 0.24830 | 15.50 | 0.0002 |
| Spend_save_quaterly_ratio | 0.06974 | 0.01024 | 0.74399 | 46.43 | <.0001 |

Bounds on condition number: 27.497, 202.93

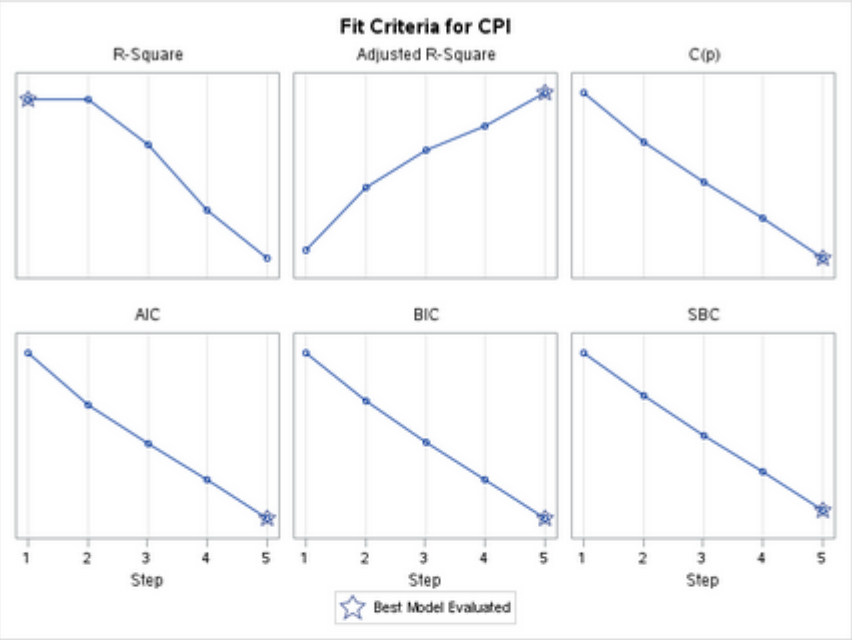
Backward Elimination: Step 6

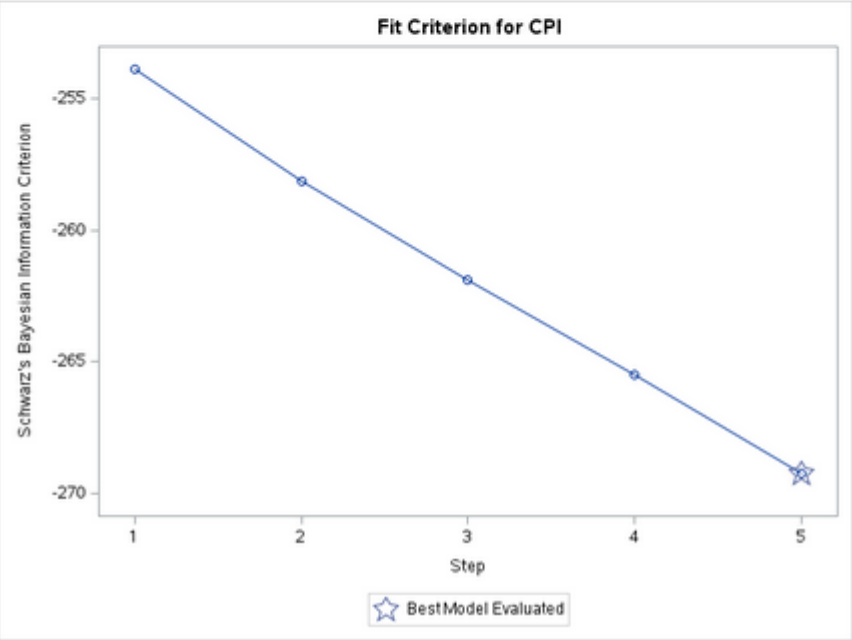
| Statistics for Removal DF = 1,64 | | | | |
|---|---|---|---|---|
| Variable | Partial R-Square | Model R-Square | F Value | Pr > F |
| Entertainment | 0.0450 | 0.9063 | 59.20 | <.0001 |
| Education | 0.0140 | 0.9374 | 18.38 | <.0001 |
| Clothing_and_shopping | 0.0118 | 0.9396 | 15.50 | 0.0002 |
| Spend_save_quaterly_ratio | 0.0353 | 0.9161 | 46.43 | <.0001 |

All variables left in the model are significant at the 0.1000 level.

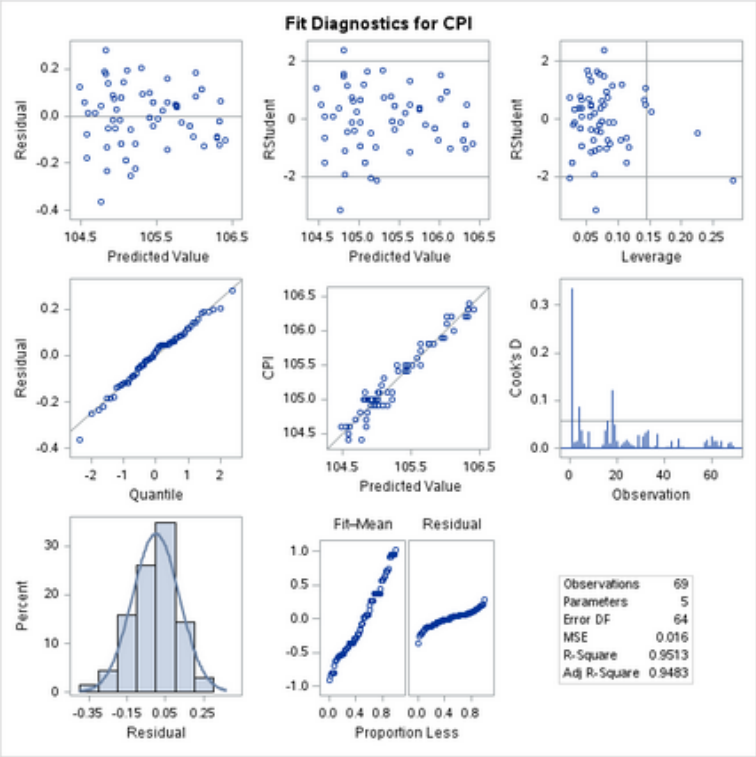
| Summary of Backward Elimination | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Step | Variable Removed | Label | Number Vars In | Partial R-Square | Model R-Square | C(p) | F Value | Pr > F |
| 1 | Eating_out | Eating_out | 8 | 0.0000 | 0.9525 | 8.0002 | 0.00 | 0.9886 |
| 2 | Grocery | Grocery | 7 | 0.0000 | 0.9524 | 6.0016 | 0.00 | 0.9701 |
| 3 | Communication | Communication | 6 | 0.0003 | 0.9521 | 4.3937 | 0.41 | 0.5267 |
| 4 | Travel_Including_Leisure | Travel_Including_Leisure | 5 | 0.0005 | 0.9517 | 2.9554 | 0.59 | 0.4467 |
| 5 | Furniture_Home_Improvement | Furniture_Home_Improvement | 4 | 0.0003 | 0.9513 | 1.3699 | 0.44 | 0.5117 |

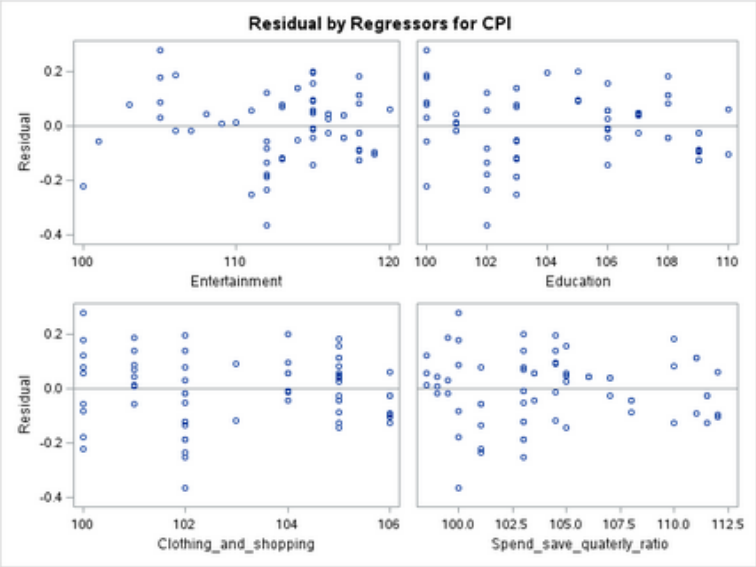
The REG Procedure
Model: MODEL2
Dependent Variable: CPI CPI




The REG Procedure
Model: MODEL2
Dependent Variable: CPI CPI




The REG Procedure
Model: MODEL3
Dependent Variable: CPI CPI

| | |
|---|---|
| Number of Observations Read | 75 |
| Number of Observations Used | 69 |
| Number of Observations with Missing Values | 6 |

Maximum R-Square Improvement: Step 1

Variable Spend_save_quaterly_ratio Entered: R-Square = 0.8661 and C(p) = 101.1808

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 1 | 18.25606 | 18.25606 | 433.27 | <.0001 |
| Error | 67 | 2.82307 | 0.04214 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 91.90309 | 0.64805 | 847.39683 | 20111.3 | <.0001 |
| Spend_save_quaterly_ratio | 0.12923 | 0.00621 | 18.25606 | 433.27 | <.0001 |

Bounds on condition number: 1, 1

The above model is the best 1-variable model found.

Maximum R-Square Improvement: Step 2

Variable Clothing_and_shopping Entered: R-Square = 0.9050 and C(p) = 54.8431

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 2 | 19.07714 | 9.53857 | 314.46 | <.0001 |
| Error | 66 | 2.00199 | 0.03033 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 85.53920 | 1.34109 | 123.40594 | 4068.34 | <.0001 |
| Clothing_and_shopping | 0.11089 | 0.02131 | 0.82107 | 27.07 | <.0001 |
| Spend_save_quaterly_ratio | 0.08058 | 0.01073 | 1.70936 | 56.35 | <.0001 |

Bounds on condition number: 4.1513, 16.607

The above model is the best 2-variable model found.

Maximum R-Square Improvement: Step 3

Variable Entertainment Entered: R-Square = 0.9374 and C(p) = 16.7114

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 3 | 19.75898 | 6.58633 | 324.29 | <.0001 |
| Error | 65 | 1.32016 | 0.02031 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 83.43388 | 1.15596 | 105.80571 | 5209.52 | <.0001 |
| Entertainment | -0.04118 | 0.00711 | 0.68184 | 33.57 | <.0001 |
| Clothing_and_shopping | 0.16070 | 0.01944 | 1.38735 | 68.31 | <.0001 |
| Spend_save_quaterly_ratio | 0.09622 | 0.00919 | 2.22716 | 109.66 | <.0001 |

Bounds on condition number: 5.161, 39.611

Variable Clothing_and_shopping Removed: R-Square = 0.8396 and C(p) = 13.9882
Variable Education Entered

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 3 | 19.80527 | 6.60176 | 336.86 | <.0001 |
| Error | 65 | 1.27386 | 0.01960 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 87.62491 | 0.70163 | 305.66773 | 15597.0 | <.0001 |
| Entertainment | -0.07407 | 0.00931 | 1.24195 | 63.37 | <.0001 |
| Education | 0.18226 | 0.02131 | 1.43365 | 73.15 | <.0001 |
| Spend_save_quaterly_ratio | 0.06784 | 0.01131 | 0.70560 | 36.00 | <.0001 |

Bounds on condition number: 14.388, 84.64

The above model is the best 3-variable model found.

Maximum R-Square Improvement: Step 5

Variable Clothing_and_shopping Entered: R-Square = 0.8513 and C(p) = 1.3898

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 4 | 20.05357 | 5.01339 | 312.86 | <.0001 |
| Error | 64 | 1.02556 | 0.01602 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 84.33993 | 1.04830 | 103.72301 | 6472.84 | <.0001 |
| Entertainment | -0.06644 | 0.00863 | 0.94870 | 59.20 | <.0001 |
| Education | 0.11236 | 0.02620 | 0.29460 | 18.38 | <.0001 |
| Clothing_and_shopping | 0.09246 | 0.02349 | 0.24830 | 15.50 | 0.0002 |
| Spend_save_quaterly_ratio | 0.06974 | 0.01024 | 0.74399 | 46.43 | <.0001 |

Bounds on condition number: 27.497, 202.83

The above model is the best 4-variable model found.

Maximum R-Square Improvement: Step 6

Variable Furniture_Home_Improvement Entered: R-Square = 0.8617 and C(p) = 2.8654

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 5 | 20.06061 | 4.01212 | 248.17 | <.0001 |
| Error | 63 | 1.01852 | 0.01617 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 84.40323 | 1.05731 | 103.02395 | 6372.50 | <.0001 |
| Furniture_Home_Improvement | 0.00767 | 0.01162 | 0.00704 | 0.44 | 0.5117 |
| Entertainment | -0.07587 | 0.01671 | 0.33320 | 20.61 | <.0001 |
| Education | 0.12025 | 0.02891 | 0.27973 | 17.30 | <.0001 |
| Clothing_and_shopping | 0.08254 | 0.02797 | 0.14074 | 8.71 | 0.0044 |
| Spend_save_quaterly_ratio | 0.07275 | 0.01125 | 0.67641 | 41.84 | <.0001 |

Bounds on condition number: 33.167, 444.45

The above model is the best 5-variable model found.

Maximum R-Square Improvement: Step 7

Variable Travel_including_Leisure Entered: R-Square = 0.8621 and C(p) = 4.3837

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 6 | 20.07016 | 3.34503 | 205.55 | <.0001 |
| Error | 62 | 1.00898 | 0.01627 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 83.31267 | 1.77582 | 35.81913 | 2201.03 | <.0001 |
| Furniture_Home_Improvement | 0.00942 | 0.01188 | 0.01023 | 0.63 | 0.4310 |
| Travel_including_Leisure | -0.00655 | 0.00855 | 0.00954 | 0.59 | 0.4467 |
| Entertainment | -0.06974 | 0.01858 | 0.22934 | 14.09 | 0.0004 |
| Education | 0.12287 | 0.02920 | 0.28805 | 17.70 | <.0001 |
| Clothing_and_shopping | 0.09037 | 0.02987 | 0.14894 | 9.15 | 0.0036 |
| Spend_save_quaterly_ratio | 0.07167 | 0.01137 | 0.64629 | 39.71 | <.0001 |

Bounds on condition number: 33.829, 706.41

The above model is the best 6-variable model found.

Maximum R-Square Improvement: Step 8

Variable Communication Entered: R-Square = 0.8624 and C(p) = 6.0018

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 7 | 20.07682 | 2.86812 | 174.55 | <.0001 |
| Error | 61 | 1.00231 | 0.01643 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 83.23078 | 1.78902 | 35.56396 | 2164.39 | <.0001 |
| Furniture_Home_Improvement | 0.00901 | 0.01195 | 0.00933 | 0.57 | 0.4539 |
| Travel_including_Leisure | -0.00739 | 0.00869 | 0.01186 | 0.72 | 0.3988 |
| Entertainment | -0.06744 | 0.01901 | 0.20669 | 12.58 | 0.0008 |
| Education | 0.11998 | 0.02969 | 0.26828 | 16.33 | 0.0002 |
| Communication | -0.00094741 | 0.00149 | 0.00666 | 0.41 | 0.5267 |
| Clothing_and_shopping | 0.09286 | 0.03027 | 0.15465 | 9.41 | 0.0032 |
| Spend_save_quaterly_ratio | 0.07274 | 0.01155 | 0.65162 | 39.66 | <.0001 |

Bounds on condition number: 34.431, 850.97

The above model is the best 7-variable model found.

Maximum R-Square Improvement: Step 9

Variable Grocery Entered: R-Square = 0.8625 and C(p) = 8.0002

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 8 | 20.07684 | 2.50960 | 150.23 | <.0001 |
| Error | 60 | 1.00229 | 0.01670 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 83.14480 | 2.91172 | 13.62114 | 815.40 | <.0001 |
| Furniture_Home_Improvement | 0.00901 | 0.01205 | 0.00934 | 0.56 | 0.4576 |
| Travel_including_Leisure | -0.00741 | 0.00878 | 0.01188 | 0.71 | 0.4024 |
| Entertainment | -0.06759 | 0.01959 | 0.19883 | 11.90 | 0.0010 |
| Grocery | 0.00130 | 0.03461 | 0.00002364 | 0.00 | 0.9701 |
| Education | 0.11976 | 0.03049 | 0.25767 | 15.42 | 0.0002 |
| Communication | -0.00094349 | 0.00150 | 0.00657 | 0.39 | 0.5328 |
| Clothing_and_shopping | 0.09292 | 0.03056 | 0.15446 | 9.25 | 0.0035 |
| Spend_save_quaterly_ratio | 0.07263 | 0.01205 | 0.60671 | 36.32 | <.0001 |

Bounds on condition number: 35.713, 1083.6

The above model is the best 8-variable model found.

Maximum R-Square Improvement: Step 10

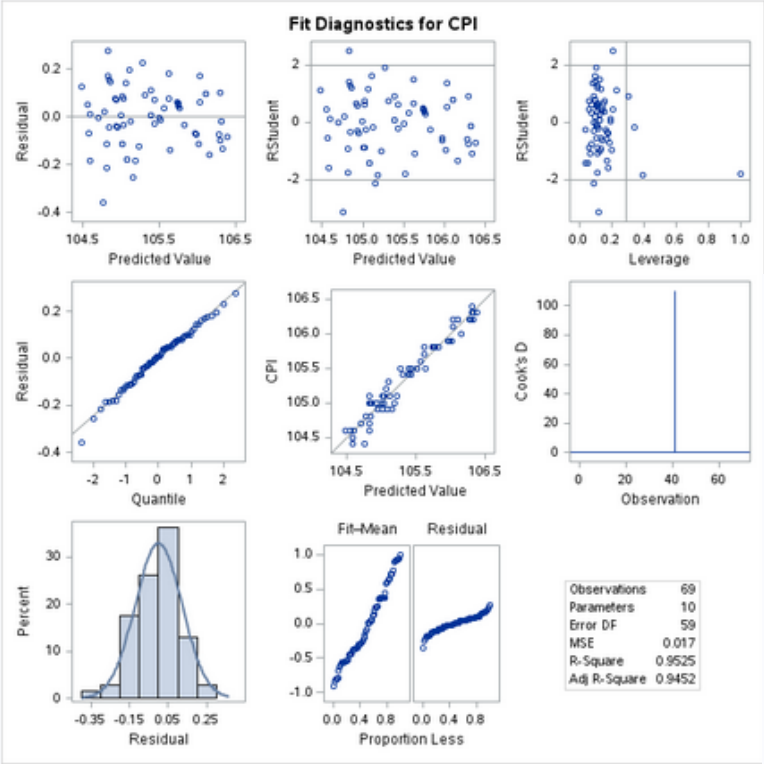
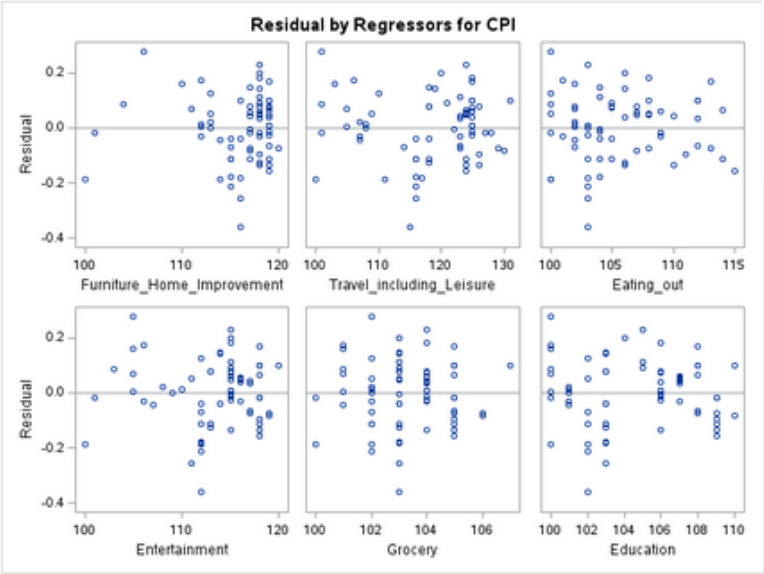
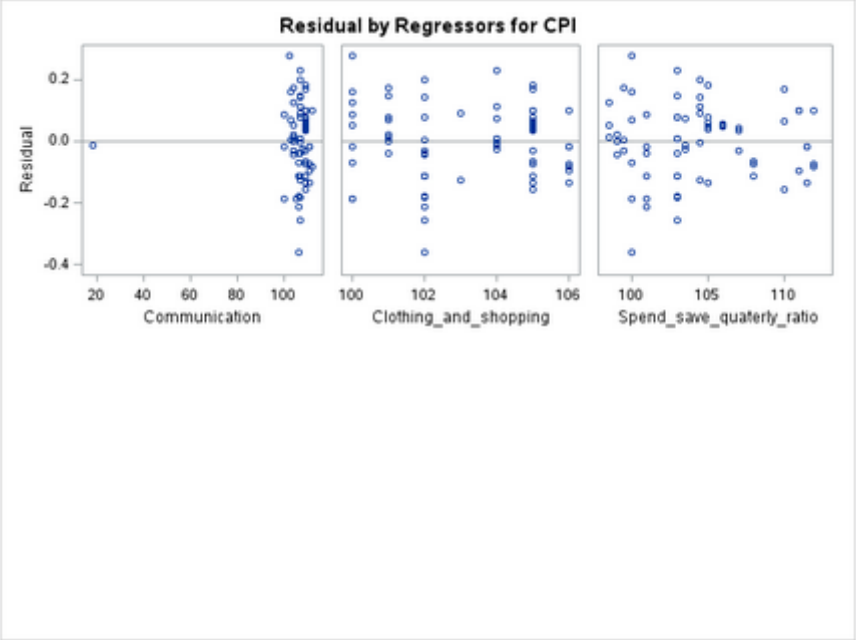
Variable Eating_out Entered: R-Square = 0.8626 and C(p) = 10.0000

| Analysis of Variance | | | | | |
|---|---|---|---|---|---|
| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model | 9 | 20.07684 | 2.23076 | 131.31 | <.0001 |
| Error | 59 | 1.00229 | 0.01699 | | |
| Corrected Total | 68 | 21.07913 | | | |

| Variable | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|---|---|---|---|---|---|
| Intercept | 83.13330 | 3.04430 | 12.66819 | 745.72 | <.0001 |
| Furniture_Home_Improvement | 0.00911 | 0.01382 | 0.00738 | 0.43 | 0.5125 |
| Travel_including_Leisure | -0.00750 | 0.01098 | 0.00793 | 0.47 | 0.4972 |
| Eating_out | -0.00015179 | 0.01061 | 0.00000348 | 0.00 | 0.9886 |
| Entertainment | -0.06756 | 0.01989 | 0.19602 | 11.54 | 0.0012 |
| Grocery | 0.00123 | 0.03527 | 0.00002060 | 0.00 | 0.9723 |
| Education | 0.11989 | 0.03200 | 0.23849 | 14.04 | 0.0004 |
| Communication | -0.00094288 | 0.00152 | 0.00656 | 0.39 | 0.5367 |
| Clothing_and_shopping | 0.09299 | 0.03123 | 0.15059 | 8.86 | 0.0042 |
| Spend_save_quaterly_ratio | 0.07273 | 0.01410 | 0.45186 | 26.60 | <.0001 |

Bounds on condition number: 38.873, 1478

The above model is the best 9-variable model found.

No further improvement in R-Square is possible.

The REG Procedure
Model: MODEL3
Dependent Variable: CPI CPI